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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Really cool team name
Selected Domain	Domain 1: Aiversecity, Future-Ready Workforce
Selected SDG Track (1–17)	Main: 4 Secondary: 8
Project / Solution Name	Unbundle
Name (Leader)	Amina Zaatir
Email & Phone (Leader)	za.amina2005@gmail.com +216 29 459 029
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

About 45% of Malaysian workers have at least 40% of their tasks that today's AI could do (ISIS Malaysia and World Bank, July 2025). Yet most people hear only a headline: "AI will change your job." Nobody tells them which of their own tasks are affected, when, or what to do next.

The people most at risk are clerical, administrative and customer-service workers, often mid-career, and graduates choosing a first job. They have the least access to personal career guidance, and generic advice such as "learn to code" ignores the tasks and skills they already have. Our core idea is that jobs do not disappear, they unbundle: some tasks move to AI, and what is left forms a new role.

This matters in three ways. Economically, reskilling money, including HRD Corp-claimable training, is wasted when it goes to skills AI will soon cover. Socially, the cost of displacement falls on the workers least able to absorb it. For sustainability, Malaysia is flagged as off track on SDG 4 (quality education and skills) in the Sustainable Development Report 2026 (SDSN), and advice that ignores the SDGs misses the green, health and care jobs the country needs. Workers need a clear, honest, personal answer, not a prediction they cannot check.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Unbundle is a web app that takes a worker's real week of tasks and skills and answers three questions: what could AI do, which jobs fit what is left, and what should I learn.

Forecast: each task gets an estimated date range for when AI could do it, under three scenarios (slowdown, normal pace, fast pace), using METR's published AI time-horizon research, Anthropic Economic Index usage data and O*NET task data.

Map: jobs are ranked by how well they fit the tasks and skills left in the person's week, with an optional "market + planet" ranking that adds points for helping SDGs on which Malaysia is off track (SDSN).

Route: the courses that close the skill gap, with hours, cost and HRD Corp status, plus the skills not worth investing in. A consent-based mentor match pairs experienced and early-career workers who can swap skills.

How AI is used: the forecast and ranking are transparent, rule-based calculations, so every number carries a source, a date and an “Estimate” label. A large language model (Qwen3-32B-AWQ, on a Runpod endpoint) is used in two places. Offline, it drafted job-to-SDG links, which are labelled estimates pending team review. At runtime, an optional box lets a worker describe their week in their own words. The model may only pick tasks and skills from our own lists, must quote the worker’s words, and every pick is checked by code and confirmed by the worker. If the model is unavailable, plain keyword matching takes over. The text is not saved. The model never produces dates, scores or rankings.

Built with what we learned in the GenAI and cloud deployment workshops.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Sponsor-provided platform: Yes, Runpod (public endpoint, Qwen3-32B-AWQ) for offline SDG-link drafting and the optional free-text intake and Replit for hosting.
Front end: React 19, Vite 8, React Router 7 (JavaScript). Plain CSS, no UI library.
Back end: Node.js 22, Express 5 REST API (/api/forecast, /match, /route, /mentor, /tasks, /skills, /sources, /refresh, /intake)
Engine: plain JavaScript modules for forecast, match, route, mentor and skills; 111 automated tests (Node's built-in test runner).
Data / database: Versioned CSV snapshots in Git (data/snapshots/YYYY-MM-DD) loaded into memory with csv-parse. Three snapshots so far (2026-03-24, 2026-10-20, 2026-10-23: 12 jobs, 200 tasks, 457 skills, 96 courses, 549 SDG links). The Refresh button also re-checks METR's public benchmark file live, validates it, and saves a new dated snapshot, or falls back to the last saved one if the check fails.
AI: Qwen3-32B-AWQ (open model, Apache 2.0) through a Runpod public endpoint: offline, to draft SDG links, and at runtime only for the optional free-text intake. Every output is validated against our own task and skill lists and confirmed by the user, with a keyword-matching fallback if the model is unavailable. Not used for any forecast, score or ranking.

Hosting: https://unbundle.replit.app/

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

1) Empathise and define: we started from two sample personas built from real O*NET task lists: Aisyah, a 45-year-old bookkeeping clerk, and Daniel, a 22-year-old graduate with no job yet. They are synthetic, not real people. Both ask the same three questions: which tasks, when, and what next.
2) Ideate and prototype: we chose a "week bar" that shows a week of tasks turning into AI-doable and still-human hours, and one clear route to a new job. We first built a mock engine with the same response shape as the real one, so design and engine work could happen in parallel, then swapped in the real engine.

3) Test: we ran a heuristic (Nielsen) review and fixed what it found: plain-words "estimate" notes, undo for removed tasks, a warning for unrealistic hours, text that does not rely on colour, a phone layout, and screen-reader labels. Testing with real users is planned for Phase 2.

Deployment: https://unbundle.replit.app/ deployed on Replit Autoscale

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used. The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Target 8.5: "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number? e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	45% of Malaysian workers have at least 40% of their tasks that today's AI could do [1]. Malaysia is flagged off track on SDG 4 [2]. In our own data, only 66% (102 of 154) of the skills missing for 7 of our 12 target jobs have any course in our catalogue, and 0 of 96 courses have a checked HRD Corp claim status [3].
Core Measurable KPI *The specific unit you will track e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Route coverage: % of the skills a worker is missing for a matched job (7 of our 12 roles) that have at least one course with stated hours and cost. Computed automatically from each data snapshot. Also tracked: % of courses whose HRD Corp claim status has been checked. HRD Corp status checked: % of courses whose HRD Corp claim status has been checked. Baseline: 0% (0 of 96).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation e.g., "20% improvement in computational resource use."*	Raise route coverage by 29%, from 66% to 85% of missing skills (29 more skills with a costed course). Check the HRD Corp claim status of at least 50% of courses (48 of 96), up from 0%.

[1] ISIS Malaysia and World Bank, Novel AI Technologies and the Future of Work in Malaysia, policy brief, July 2025
[2] SDSN, Sustainable Development Report 2026
[3] Unbundle data snapshot 2026-10-23

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✓ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type	License
React, React DOM, React Router, Vite	code	MIT

Express, csv-parse, js-yaml, concurrently	code	MIT
dotenv	code	BSD-2-Clause
Qwen3-32B-AWQ (via Runpod endpoint)	model	Apache 2.0
O*NET 31.0 (US Dept. of Labor)	dataset	CC BY 4.0
ESCO (European Commission)	dataset	CC BY 4.0
Anthropic Economic Index	dataset	CC BY 4.0
METR time-horizon benchmark data	dataset	Publicly available data published by METR (metr.org); used with attribution, per METR's repository terms.
UN SDG API / SDG targets (UN Statistics Division)	dataset	Free with citation
SDSN Sustainable Development Report 2026	dataset	used with SDSN's email permission (21 Oct 2026)
US BLS employment projections	dataset	US government public data
DOSM Malaysia statistics	dataset	CC BY 4.0

Microsoft Learn Catalog API, Coursera public catalogue	dataset	their API terms of use
Aurora SDG queries	dataset	CC BY 4.0
ISIS Malaysia / World Bank policy brief	research	cited, not copied

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25/10/2026

Appendix A: System Block Diagram

